Use Case: Connect 4 Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: Allow the player to access the main Connect 4 game interface and select options such as starting a new game, viewing the leaderboard, or joining a queue for matchmaking.

Preconditions:

- The player is logged in and has access to their profile.
- The Connect 4 game module is available.

Trigger: The player navigates to the Connect 4 game section from the Player Profile Home Menu.

Scenario:

1. The player selects “Connect 4” from the Player Profile Home Menu.
2. The system loads the Connect 4 Home Menu.
3. The player sees options to view the Connect 4 leaderboard, or join the matchmaking queue.
4. The player selects their desired option (e.g., “Join Queue”).
5. The system directs the player to the chosen option.

Post Conditions: The player has accessed their chosen option within the Connect 4 module.

Exceptions:

- **E1:** The Connect 4 module is unavailable. The system shows an error message and redirects the player to the Player Profile Home Menu.

Priority: High

When Available: Upon player login and successful navigation to Connect 4.

Frequency of Use: Frequent, as this is the entry point for Connect 4-related activities.

Channel to Actor: User Interface (UI)

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues:

- Should the Connect 4 Home Menu include additional options, such as game settings or tutorials? Or play with AI? Or local multiplayer?

Use Case: Connect 4 Game Queue

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to enter a queue for matchmaking in order to find an opponent for a Connect 4 game.

Preconditions:

- The player has accessed the Connect 4 Home Menu.
- The matchmaking service is active.

Trigger: The player selects the option to join the Connect 4 game queue.

Scenario:

1. The player selects “Join Queue” from the Connect 4 Home Menu.
2. The system adds the player to the Connect 4 game queue.
3. The system continuously searches for a compatible opponent.
4. If an opponent is found, the system notifies the player and initiates the Connect 4 Game Instance.

Post Conditions: The player has either joined a match or is informed that no match is currently available.

Exceptions:

- **E1:** No players are available in the queue. The system displays a “No Match Found” message.
- **E2:** Matchmaking service is down. The system shows an error message and returns the player to the Connect 4 Home Menu.

Priority: High

When Available: Anytime the player chooses to enter the queue.

Frequency of Use: Moderate, dependent on player activity and matchmaking wait times.

Channel to Actor: UI notification (possibly with a “Searching for opponent…” indicator)

Secondary Actors: Matchmaking service

Channels to Secondary Actors: System-to-system communication with matchmaking service

Open Issues:

- What criteria are used to find a compatible opponent (e.g., skill level)?
- Should there be an estimated wait time shown to the player?

Use Case: Connect 4 Game Instance

Iteration: 1

Primary Actor: Player

Goal In Context: Allow the player to participate in a Connect 4 game against an opponent.

Preconditions: A match has been successfully created, and an opponent has been assigned.

Trigger: The system initiates the game instance upon finding an opponent for the player in the queue.

Scenario:

1. The system matches the player with an opponent and starts the game instance.
2. The player is notified that the game is starting and is presented with the Connect 4 game board.
3. The player and the opponent take turns making moves until the game ends (either by winning, losing, or a draw).
4. The system records the game result and updates the leaderboard.
5. The player is shown the game result screen (victory, defeat, or draw).
6. The player has the option to return to the Connect 4 Home Menu or view the leaderboard.

Post Conditions: The game results are recorded, and the player is directed to the game aftermath options.

Exceptions:

- **E1:** Opponent disconnects. The player is informed, and the game ends prematurely.
- **E2:** Connection loss. The system saves the game state temporarily or informs the player of the disconnection.

Priority: High

When Available: Once matchmaking is complete.

Frequency of Use: Dependent on player's success in finding a match.

Channel to Actor: UI with real-time updates

Secondary Actors: Opponent player

Channels to Secondary Actors: Network communication for real-time game moves

Open Issues:

- Should there be an option to rematch with the same opponent?
- How is the game state handled if a player disconnects?

Use Case: No Match Found

Iteration: 1

Primary Actor: Player

Goal In Context: Notify the player that no match is currently available in the queue and provide options to retry or exit.

Preconditions: The player has joined the Connect 4 Game Queue.

Trigger: The system fails to find an opponent within a defined timeout period or due to low player availability.

Scenario:

1. The player joins the Connect 4 Game Queue.
2. The system continuously searches for an opponent.
3. After a specified timeout period, the system fails to find a match.
4. The system displays a “No Match Found” notification to the player.
5. The player is given options to retry joining the queue or to return to the Connect 4 Home Menu.

Post Conditions: The player is informed that no match was found and can choose to retry or exit.

Exceptions:

- **E1:** System experiences an error in retrieving player queue status. An error message is shown, and the player is redirected to the Connect 4 Home Menu.

Priority: Medium

When Available: During periods of low player activity or when the matchmaking queue has a timeout.

Frequency of Use: Occasional, depending on player availability and queue demand.

Channel to Actor: UI notification

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues:

- How long should the system wait before displaying the “No Match Found” message?
- Should the player receive an estimated wait time or indication of current queue length?

Use Case: Connect Four Leaderboard

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to view the leaderboard for Connect Four, displaying essential rankings and game-related statistics of top players.

Preconditions:

- The player has accessed the Connect Four Home Menu.
- Leaderboard data is available and up-to-date.

Trigger: The player selects the “Leaderboard” option from the Connect Four Home Menu.

Scenario:

1. The player navigates to the Connect Four Home Menu.
2. The player selects “Leaderboard.”
3. The system retrieves the latest leaderboard data for Connect Four.
4. The system displays the leaderboard, showing the following statistics for each player:
    - **Rank**: The player’s position based on win rate or score.

- **Username**: The display name of each ranked player.
- **Total Wins**: The total number of games the player has won.
- **Win Rate**: The percentage of games won out of total games played.
- **Longest Win Streak**: The highest number of consecutive wins achieved.
- **Games Played**: The total number of games the player has participated in.
- **Last Played Date**: The date of the player's most recent game.
- **Moves to Win (Average)**: The average number of moves the player needed to win a game.

5. The player can scroll through or search the leaderboard to find specific players or rankings.
6. The player can filter the leaderboard based on categories like "Top Wins" or "Highest Win Rate."
7. The player can return to the Connect Four Home Menu or view their own profile from the leaderboard page.

Post Conditions: The player has successfully viewed the Connect Four leaderboard and can navigate back to other Connect Four options.

Exceptions:

- **E1:** The leaderboard data fails to load due to a network issue. The system displays an error message and provides an option to retry or return to the Connect Four Home Menu.
- **E2:** The leaderboard service is temporarily unavailable. The system informs the player and directs them to try again later or return to the Connect Four Home Menu.

Priority: Medium

When Available: Anytime the player is in the Connect Four Home Menu.

Frequency of Use: Occasional, based on the player's interest in checking rankings and scores.

Channel to Actor: UI display on the leaderboard screen

Secondary Actors: Leaderboard database/service

Channels to Secondary Actors: System-to-database communication to retrieve leaderboard information

Open Issues:

- Should the leaderboard allow additional filtering options (e.g., friends, region)?
- How frequently should the leaderboard data be refreshed for accuracy?